Tutorial #5: Lecture 2 Part 2

Section 2.4:

2). a). $2^{n-1} \Rightarrow a_8$ is $2^{8-1} = 128$

b). $7 \Rightarrow a_8 = 7$

c). $1+(-1)^n \Rightarrow a_8 = 1+(-1)^8 = 1+1 = 2$

d). $-(-2)^n \Rightarrow a_8 = -(-2)^8 = -256$

4). a). $(-2)^n$?: $a_0 = (-2)^0 = 1$, $a_1 = (-2)^1 = -2$, $a_2 = (-2)^2 = 4$, $a_3 = (-2)^3 = -8$

b). 3? : $a_0 = 3$, $a_1 = 3$, $a_2 = 3$, $a_3 = 3$

c). $7+4^n$?: $a_0 = 7+4^0 = 8$, $a_1 = 7+4^1 = 11$, $a_2 = 7+4^2 = 15$, $a_3 = 7+4^3 = 71$

d). $2^n+(-2)^n$?: $a_0 = 2^0+(-2)^0 = 2$, $a_1 = 2^1+(-2)^1 = 0$

$a_2 = 2^2+(-2)^2 = 8$, $a_3 = 2^3+(-2)^3 = 0$

12). Show that the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = -3a_{n-1} + 4a_{n-2}$ if

a). $a_n = 0$? $\quad -3a_{n-1} + 4a_{n-2} = -3 \cdot 0 + 4 \cdot 0 = 0 = a_n$ ✓

b). $a_n = 1$? $\quad -3a_{n-1} + 4a_{n-2} = -3 \cdot 1 + 4 \cdot 1 = 1 = a_n$ ✓

c). $a_n = (-4)^n$? $\quad -3a_{n-1} + 4a_{n-2} = -3(-4)^{n-1} + 4(-4)^{n-2}$

$= (-4)^{n-2}\left[-3(-4) + 4\right]$

$= (-4)^{n-2}(12+4) = (-4)^{n-2} \cdot (-4)^2 = (-4)^n = a_n$ ✓

d). $a_n = 2(-4)^n + 3$? $\quad -3a_{n-1} + 4a_{n-2} = -3\left[2(-4)^{n-1} + 3\right] + 4\left[2(-4)^{n-2} + 3\right]$

$= \underline{-6(-4)^{n-1}} - 9 + \underline{8(-4)^{n-2}} + 12 = (-4)^{n-2}\left[\underbrace{-6(-4)}_{24} + 8\right] + 3$

$$= 32(-4)^{n-2} + 3 = 2 \cdot \underbrace{16}_{(-4)^2}(-4)^{n-2} + 3 = 2(-4)^n + 3 = a_n \checkmark$$

30). $S = \{1, 3, 5, 7\}$

a). $\sum_{j \in S} j = 1 + 3 + 5 + 7 = 16$

b). $\sum_{j \in S} j^2 = 1^2 + 3^2 + 5^2 + 7^2 = 84$

c). $\sum_{j \in S} (1/j) = (1/1) + (1/3) + (1/5) + (1/7) = 176/105$

d). $\sum_{j \in S} 1 = 1 + 1 + 1 + 1 = 4$

32). a). $\sum_{j=0}^{8} (1 + (-1)^j) = (1 + (-1)^0) + (1 + (-1)^1) + (1 + (-1)^2 + (1 + (-1)^3) +$
$(1 + (-1)^4) + (1 + (-1)^5) + (1 + (-1)^6) + (1 + (-1)^7) +$
$(1 + (-1)^8 = 2 + 0 + 2 + 0 + 2 + 0 + 2 + 0 + 2 = 10$

b). $\sum_{j=0}^{8} (3^j - 2^j) = \left(\sum_{j=0}^{8} 3^j\right) - \left(\sum_{j=0}^{8} 2^j\right)$

Using Table 2, $\sum_{k=0}^{n} ar^k \ (r \neq 0) = \dfrac{ar^{n+1} - a}{r - 1}, \ r \neq 0$

(Geometric progression Def. 2)

$$\sum_{j=0}^{8} 3^j = \frac{3^{8+1} - 1}{3 - 1} = 9841$$

$$\sum_{j=0}^{8} 2^j = \frac{2^{8+1} - 1}{2 - 1} = 511$$

$$\therefore \sum_{j=0}^{8} (3^j - 2^j) = 9841 - 511 = 9330 \ \#.$$

34). a). $\sum_{i=1}^{3}\sum_{j=1}^{2}(i-j) = \underbrace{[(1-1)+(1-2)]}_{i=1} + \underbrace{[(2-1)+(2-2)]}_{i=2} + \underbrace{[(3-1)+(3-2)]}_{i=3} = 3$

b). $\sum_{i=0}^{3}\sum_{j=0}^{2}(3i+2j) = \underbrace{[(0+0)+(0+2)+(0+4)]}_{i=0} + \underbrace{[(3+0)+(3+2)+(3+4)]}_{i=1}$

$+ \underbrace{[(6+0)+(6+2)+(6+4)]}_{i=2} + \underbrace{[(9+0)+(9+2)+(9+4)]}_{i=3}$

$=$

c). $\sum_{i=1}^{3}\sum_{j=0}^{2} j = \underbrace{(0+1+2)}_{i=1} + \underbrace{(0+1+2)}_{i=2} + \underbrace{(0+1+2)}_{i=3} = 9$

d). $\sum_{i=0}^{2}\sum_{j=0}^{3} i^2 j^3 = (\underbrace{0}_{0^2\times 0^3}+\underbrace{0}_{0^2\times 1^3}+\underbrace{0}_{0^2\times 2^3}+\underbrace{0}_{0^2\times 3^3}) + (\underbrace{0}_{1^2\times 0^3}+\underbrace{1}_{1^2\times 1^3}+\underbrace{8}_{1^2\times 2^3}+\underbrace{27}_{1^2\times 3^3}) + (\underbrace{0}_{2^2\times 0^3}+\underbrace{4}_{2^2\times 1^3}+\underbrace{32}_{2^2\times 2^3}+\underbrace{108}_{4\times 27\ (2^2\times 3^3)}) = 180$ #

40). Find $\sum_{k=99}^{200} k^3$

Using Table 2 $\sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4}$

$$\sum_{k=1}^{200} k^3 = \frac{200^2(200+1)^2}{4} = \frac{200^2(201)^2}{4} = 404{,}010{,}000$$

$$\sum_{k=1}^{98} k^3 = \frac{98^2(98+1)^2}{4} = \frac{98^2(99)^2}{4} = 23{,}532{,}201$$

$$\sum_{k=99}^{200} k^3 = \sum_{k=1}^{200} k^3 - \sum_{k=1}^{98} k^3 = 380{,}477{,}799$$ #.

Section 2.5

1. a). the negative integers $\Rightarrow -1, -2, -3, -4, \ldots$
   Countably infinite.

   b). the even integer $\Rightarrow 0, 2, -2, 4, -4, 6, -6, \ldots$
   Countably infinite.

   c). the integers less than 100 $\Rightarrow 99, 98, 97, \ldots$
   Countably infinite

   d) the real number between 0 and $\frac{1}{2}$ $\Rightarrow$ Uncountable

   e). the positive integers less than 1,000,000,000
   Countably finite.

   f). the integers that are multiples of 7 $\Rightarrow 0, 7, -7, 14, -14, \ldots$
   Countably infinite

4. a). integers not divisible by 3

   $\pm 1, \pm 2, \pm 4, \pm 5, \pm 7, \ldots$

   list these numbers in the order

   $1, -1, 2, -2, 4, -4, 5, -5, 7, -7, \ldots$

   We can have a one-to-one correspondence between the set of positive integers and this set as

   $1 \leftrightarrow 1$
   $2 \leftrightarrow -1$
   $3 \leftrightarrow 2$
   $4 \leftrightarrow -2$
   $5 \leftrightarrow 4$
   $6 \leftrightarrow -4$
   $\vdots$

b). integers divisible by 5 but not by 7

$$\begin{array}{cccccccccc} 5, & -5, & 10, & -10, & 15, & -15, & 20, & -20, & 25, & -25 \ldots \\ \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \ldots \end{array}$$

Section 2.6

1). a). 3×4 b. $\begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix}$ c. $\begin{bmatrix} 2 & 0 & 4 & 6 \end{bmatrix}$ d). 1 e) $\begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 1 & 4 & 3 \\ 3 & 6 & 7 \end{bmatrix}$

2). a) $\begin{bmatrix} 0 & 3 & 9 \\ 1 & 4 & -1 \\ 2 & -5 & -3 \end{bmatrix}$ b). $\begin{bmatrix} -4 & 9 & 2 & 10 \\ -4 & -5 & 4 & 0 \end{bmatrix}$

4). a) $$\begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \overset{-1}{0+0-1} & 1 & 0 \\ 0 & 1 & -1 \\ 1 & -2 & 1 \end{bmatrix}$$

b). $$\begin{bmatrix} 1 & -3 & 0 \\ 1 & 2 & 2 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 & 3 \\ -1 & 0 & 3 & -1 \\ -3 & 2 & 0 & 2 \end{bmatrix} = \begin{bmatrix} \overset{4}{1+3+0} & -1 & -7 & 6 \\ -7 & -5 & 8 & 5 \\ 4 & 0 & 7 & 3 \end{bmatrix}$$

c). $$\begin{bmatrix} 0 & -1 \\ 7 & 2 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} 4 & -1 & 2 & 3 & 0 \\ -2 & 0 & 3 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 0+2 & 0 & -3 & -4 & -1 \\ 24 & -7 & 20 & 29 & 2 \\ -10 & 4 & -17 & -24 & -3 \end{bmatrix}$$

8). By Definition 3, the entries of $A+B$ are $a_{ij}+b_{ij}$

and —— " —— $B+A$ are $b_{ij}+a_{ij}$

Commutativity of addition of real numbers

$$a_{ij}+b_{ij} = b_{ij}+a_{ij}$$

$$\therefore A+B = B+A \quad \#$$

26). $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

a). $A \vee B = \begin{bmatrix} 1 \vee 0 & 1 \vee 1 \\ 0 \vee 1 & 1 \vee 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

b). $A \wedge B = \begin{bmatrix} 1 \wedge 0 & 1 \wedge 1 \\ 0 \wedge 1 & 1 \wedge 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

c). $A \odot B = \begin{bmatrix} (1 \wedge 0) \vee (1 \wedge 1) & (1 \wedge 1) \vee (1 \wedge 0) \\ (0 \wedge 0) \vee (1 \wedge 1) & (0 \wedge 1) \vee (1 \wedge 0) \end{bmatrix}$

$= \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

28).

$A \odot B = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \odot \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) \vee (0 \wedge 1) \vee (1 \wedge 1) & 0 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}$

$A \odot B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}$